

Answer ALL questions.

Write your answers in the spaces provided.

- 1 The photograph shows elephant grass (*Pennisetum purpureum*).



(Source: © ASHLEY COOPER/SCIENCE PHOTO LIBRARY)

Elephant grass grows well in the nutrient-poor soils in dry areas in East Africa and is widely used as food for cattle.

Its use as a source of fibres has been studied.

- (a) The strength of plant fibres is partly due to the arrangement of cellulose microfibrils in the cell walls of these fibres.

- (i) Name **two** types of cell found in plant fibres.

(2)

1

2



- (ii) Fibres have cell walls with secondary thickening of lignin.

Describe **two** ways in which secondary thickening affects the properties of the cell.

(2)

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- (b) Soaking plant fibres in sodium hydroxide causes them to swell and so increases their strength.

The effect of length of time of soaking on the strength of elephant grass fibres was investigated.

The breaking strength of the fibres after this soaking was determined to calculate the tensile strength, in Newtons per square millimetre (Nmm^{-2}).

- (i) Name the independent variable in this investigation.

(1)

- (ii) After soaking, the diameter of each fibre was measured.

Describe how fibre diameter could be measured accurately using a microscope.

(3)

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P 7 1 9 3 6 R A 0 3 2 0

(iii) Give **two** reasons why fibre diameter was measured.

(2)

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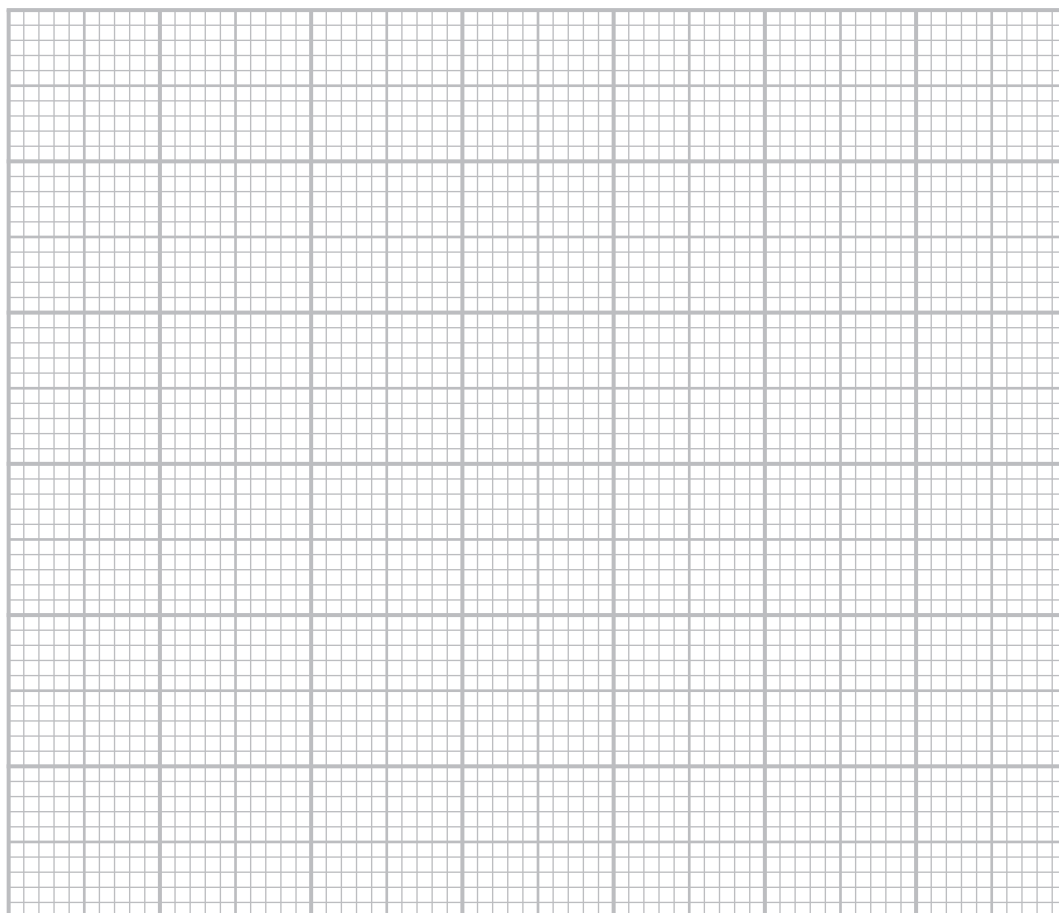


(c) Some of the results of this investigation are shown in the table.

Soaking time / hours	Tensile strength / Nmm^{-2}
0	38
3	60
6	86
12	88
18	90
24	90

- (i) Plot a suitable graph to show the effect of soaking time on tensile strength. Join the points with straight lines.

(4)



(ii) Describe the results of this investigation.

(3)

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(Total for Question 1 = 17 marks)

